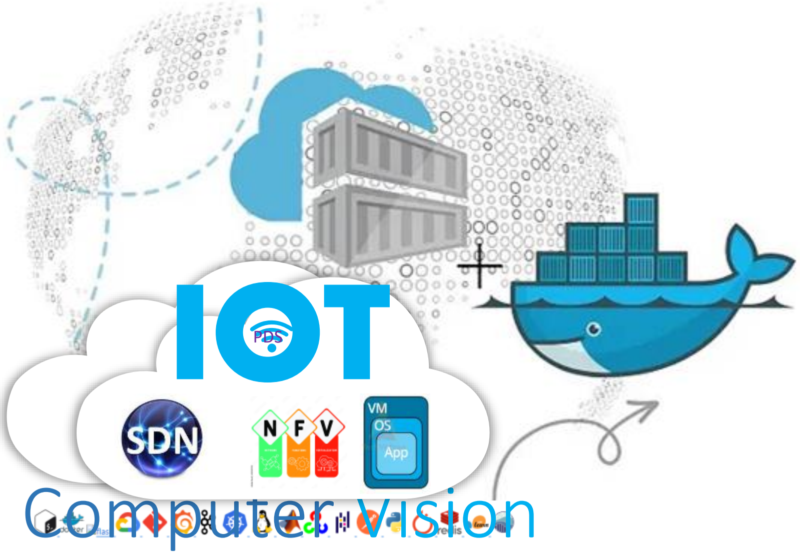

Kolla Ansible OpenStack Installation (Ubuntu server 22.04)

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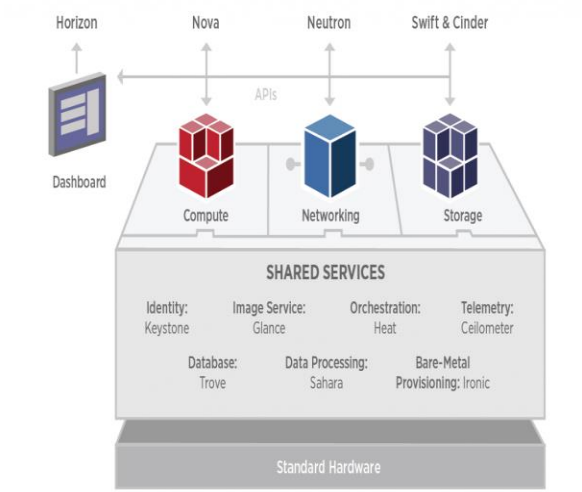
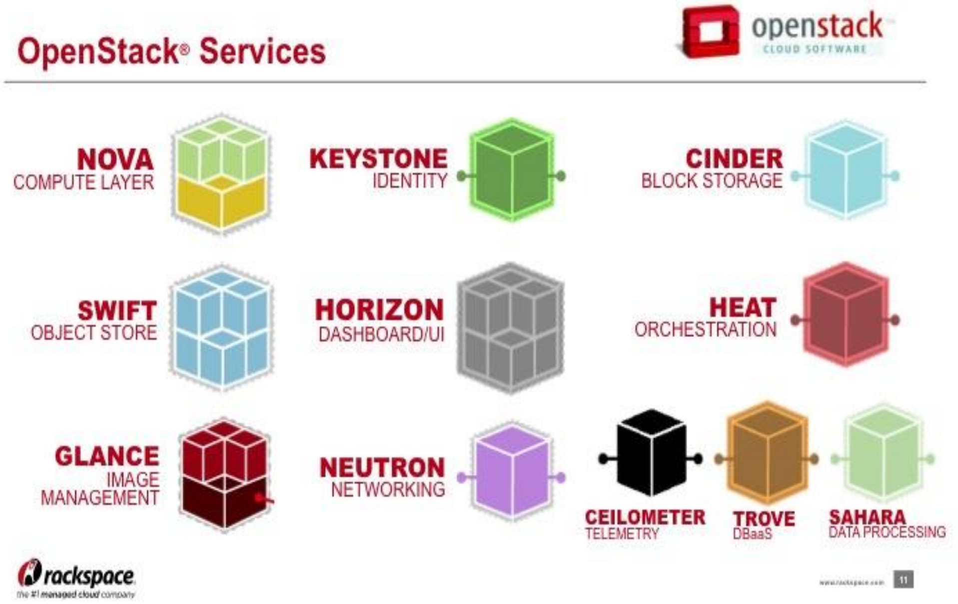
https://github.com/siagianp
https://www.youtube.com/@AmelOline/videos
https://github.com/amelcharolinesgn2/IoT_simulator-mqtt-NodeRed
https://github.com/amelcharolinesgn2/Cloud-Infrastructures
https://github.com/amelcharolinesgn2/ClouD-Infrastructure-SISKA-2025

Arsitektur Openstack





Openstack Services

Kolla Ansible melakukan deploy container untuk project Openstack berikut:

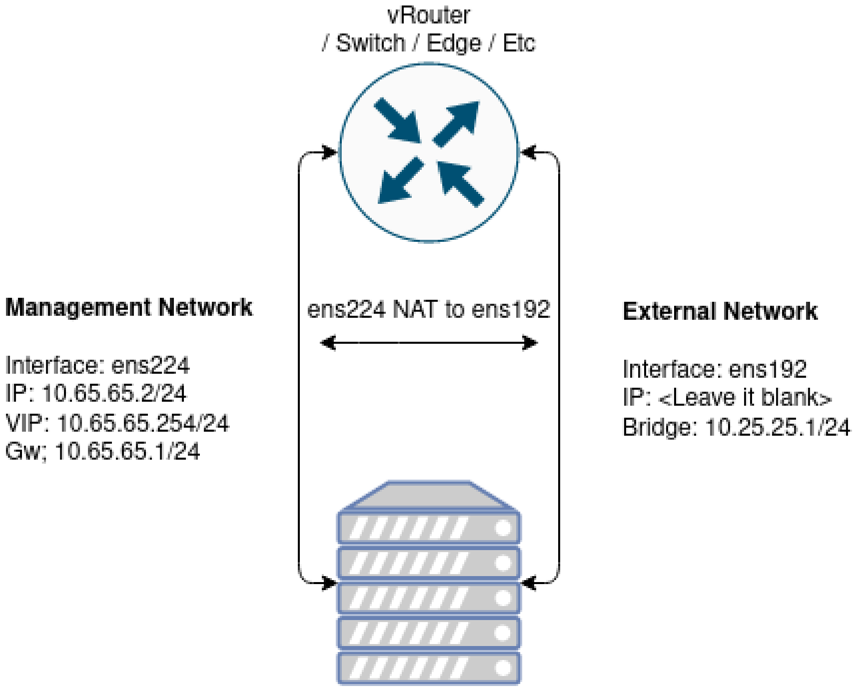
Aodh
Barbican
Bifrost
Blazar
Ceilometer
Cinder
CloudKitty
Congress
Designate
Freezer
Glance
Heat
Horizon
Ironic
Karbor
Keystone
Kuryr
Magnum
Manila
Mistral
Murano

Infrastructure components

Service Openstack berikut:

- ✓ Ceph implementation for Cinder, Glance and Nova.
- ✓ Collectd, Telegraf, InfluxDB, Prometheus, and Grafana for performance monitoring.
- ✓ Elasticsearch and Kibana to search, analyze, and visualize log messages.
- ✓ Etcd a distributed reliable key-value store.
- ✓ Fluentd as an open source data collector for unified logging layer.
- ✓ Gnocchi A time-series storage database.
- ✓ HAProxy and Keepalived for high availability of services and their endpoints.
- ✓ MariaDB and Galera Cluster for highly available MySQL databases.
- ✓ Memcached a distributed memory object caching system.
- ✓ MongoDB as a database back end for Panko.
- ✓ Open vSwitch and Linuxbridge backends for Neutron.
- ✓ RabbitMQ as a messaging backend for communication between services.
- ✓ Redis an in-memory data structure store.

Lab: Kolla Ansible All in One Openstack



Ubuntu 20.04
Openstack

Requirement

Kali ini kami akan menggunakan spesifikasi mesin dan os seperti berikut, silakan dapat diikuti atau dapat disesuaikan dengan kebutuhan Anda:

- ✓ 8 CPU
- ✓ 8 GB RAM
- ✓ 100 GB Disk
- ✓ Ubuntu 20.04
- ✓ 2 Network Interface

Lab: Kolla Ansible All in One Openstack (Cons't)

2 NIC

Membutuhkan minimal 2 network interface yang nantinya akan digunakan sebagai:

1. Management Network

Interface: ens224 IP: 10.65.65.2/24 VIP: 10.65.65.254/24 Gw; 10.65.65.1/24

2. External Network

Interface: ens192 IP: kosongkan Bridge: 10.25.25.1/24

Management Network digunakan sebagai media komunikasi antar service yang menjalankan openstack, sedangkan External Network digunakan service neutron untuk berkomunikasi antara network internal (private) pada instances dan host (internet/publik), untuk interface ini diharuskan berstatus aktif (up) dan tidak dialokasikan IP pada interfacenya.

Konfigurasi Network

Saran...!!!!
Menggunakan ip statis pada setiap interface untuk menghindari perubahan IP address / routing pada saat host restart.
Network manager saya menggunakan netplan, berikut adalah konfigurasi interfacenya:

```
vi /etc/netplan/01-iface.yaml
...
# This is the network config written by 'subiquity'
network:
 ethernets:
  ens224:
   dhcp4: false
   addresses: [10.65.65.2/24]
   nameservers:
     addresses: [1.1.1.1,1.0.0.1]
  ens192: {}
 version: 2
 ...
  netplan apply
```

Instalasi dependensi

Menggunakan media virtual environment yang disediakan python, adapun dependensi yang perlu diinstall adalah berikut:

root@administrator:/home/panda/Deploy#

Update paket manager:

 # sudo apt update

Install dependensi python3-venv:

 # sudo apt install python3-venv

Buat sebuah virtual environment lalu aktifkan untuk mulai menggunakannya:

- Buat folder baru pada direktori home ***panda*** denga nama **Deploy**, bisa disesuaikan dengan fungsinya :)
- mengaktifkan virtual environment dahulu sebelum memulai menggunakan perintah python

```
mkdir deploy
python3 -m venv /home/panda/Deploy
source /home/panda/Deploy/bin/activate
```

Upgrade versi pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

root@administrator:/home/panda/Deploy#

 # pip install –U pip

Install paket ansible :

Kolla ansible membutuhkan paket ansible minimal versi 4 dan mendukung hingga versi 5 keatas.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
#pip install 'ansible>=4,<6'
```

Lab: Kolla Ansible All in One Openstack (Cons't)

root@administrator:/home/panda/Deploy#

Install paket docker:

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# pip install docker
```

Install Kolla-ansible

Install kolla-ansible beserta dependensinya menggunakan python-pip

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
#pip install git+https://opendev.org/openstack/kolla-ansible@master
```

Buat directory /etc/kola

root@administrator:/home/panda/Deploy#

```
# sudo mkdir -p /etc/kolla
# sudo chown $panda:$panda /etc/kolla
```

Salin file global.yml dan password.yml ke directory /etc/kolla:

```
# cp -r /home/panda/deploy/share/kolla-ansible/etc_examples/kolla/* /etc/kolla
```

Salin file konfigurasi all-in-one:

```
# cp /path/to/venv/share/kolla-ansible/ansible/inventory/all-in-one /home/panda/
```

Tambahkan python interpreter pada konfigurasi all-in-one

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# source /home/panda/deploy/bin/activate
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Cari lokasi binary python:

```
which python
/home/panda/Deploy/bin/python
```

Tambahkan interpreter sesuai lokasi

```
vi all-in-one
...
Localhost
ansible_python_interpreter=/home/panda/deploy/bin/python
```

Install Ansible Galaxy requirements

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-ansible install-deps
```

Konfigurasi Awal

Generate Password

Password yang digunakan oleh kolla ansible disimpan di direktori **/etc/kolla/passwords.yml**

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

```
# kolla-genpwd
```

Konfigurasi global.yml

```
vi /etc/kolla/global.yml
...
kolla_base_distro: "ubuntu"
openstack_release: "master"
kolla_internal_vip_address: "10.65.65.254"
network_interface: "ens224"
neutron_external_interface: "ens192"
neutron_plugin_agent: "openvswitch"
enable_neutron_provider_networks: "yes"
nova_compute_virt_type: "kvm"
```


Simpan konfigurasi dan lanjut ke deployment !

Deployment

Kolla Ansible sudah menyediakan playbook list yang akan menginstal semua service yang sudah di set pada konfigurasi sebelumnya.

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

@ Lakukan bootstraping pada host menggunakan dependensi kolla ansible:

```
# kolla-ansible -i all-in-one bootstrap-servers
```

@ Lakukan pre-checks konfigurasi

```
# kolla-ansible -i all-in-one prechecks
```

@ Jika dari hasil check sudah tidak ada error, selanjutnya deploy server

```
# kolla-ansible -i all-in-one deploy
```

Proses deployment akan membutuhkan waktu yang agak lama, pastikan saja kalian sudah menjalankan **screen** bila menggunakan remote SSH.

Jika deployment berhasil output akhirnya kurang lebih seperti berikut:

@ Lakukan bootstraping pada host menggunakan dependensi kolla ansible:

```
...
PLAY RECAP
**************************************************
**************************************************
**************************************************
*********************************
localhost            : ok=366  changed=72
unreachable=0   **failed=0**   skipped=245
rescued=0   ignored=0
```

Lab: Kolla Ansible All in One Openstack (Cons't)

List Running OpenStack Docker Containers

Untuk mengecek service docker yang berjalan silakan menggunakan command ini:

```
# docker ps
```

OpenStack Post Deployment

Pastikan virtual environment sudah aktif sebelum menjalankan command ini:

Install Openstack Client

```
# pip install python-openstackclient python-neutronclient python-glanceclient
```

Generate AdminRC

```
# kolla-ansible post-deploy
```

Gunakan AdminRC

```
# source /etc/kolla/admin-openrc.sh
```

Jalankan script untuk inisiasi awal

Secara default kolla ansible sudah menyiapkan script inisiasi awal yang digunakan untuk create network demo didalam openstack.

```
vim /home/panda/deploy/share/kolla-ansible/init-runonce
...
# This EXT_NET_CIDR is your public network,that you want to connect to the internet via.
ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}
EXT_NET_CIDR=${EXT_NET_CIDR:-'10.25.25.0/24'}
EXT_NET_RANGE=${EXT_NET_RANGE:-'start=10.25.25.4,end=10.25.25.253'}
EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-'10.25.25.1'}
```

Lab: Kolla Ansible All in One Openstack (Cons't)

Eksekusi script

```
/home/panda/deploy/share/kolla-ansible/init-runonce
```

Aktifkan ipv4 forwarding

```
vi /etc/sysctl.conf
...
net.ipv4.ip_forward=1
net.ipv4.ip_nonlocal_bind=1
net.ipv6.ip_nonlocal_bind=1
net.unix.max_dgram_qlen=128
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.neigh.default.gc_thresh1=128
net.ipv4.neigh.default.gc_thresh2=28672
net.ipv4.neigh.default.gc_thresh3=32768
net.ipv6.neigh.default.gc_thresh1=128
net.ipv6.neigh.default.gc_thresh2=28672
net.ipv6.neigh.default.gc_thresh3=32768
```

Sysctl

```
# sysctl -p
```

Atur interface bridge

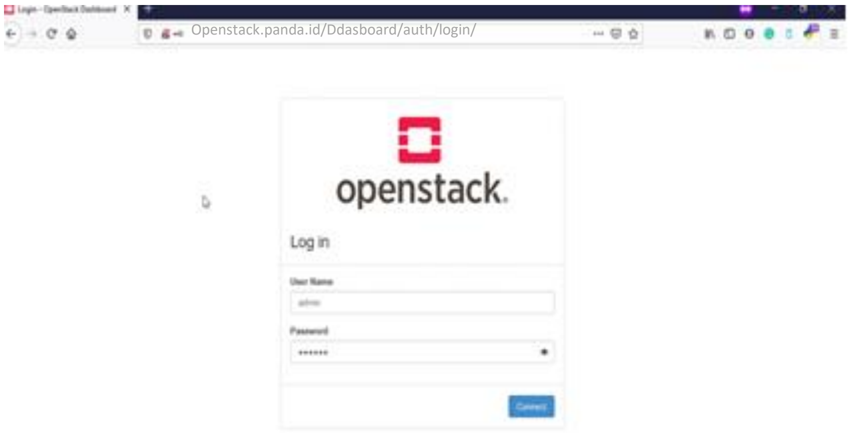
Jalankan perintah berikut ini :

```
export ENABLE_EXT_NET=${ENABLE_EXT_NET:-1}
export EXT_NET_CIDR=${EXT_NET_CIDR:-'10.25.25.0/24'}
export EXT_NET_RANGE=${EXT_NET_RANGE:-'start=10.25.25.3,end=10.25.25.253'}
export EXT_NET_GATEWAY=${EXT_NET_GATEWAY:-'10.25.25.1'}

## Akses internet ke instances
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask 255.255.255.0 up
sudo iptables -t nat -A POSTROUTING -s $EXT_NET_CIDR -o ens224 -j MASQUERADE
sudo iptables -A FORWARD -o ens224 -i br-ex -j ACCEPT
sudo iptables -A FORWARD -i ens224 -o br-ex -j ACCEPT
```

Akses dashboard dan uji konektifitas instance

Dashboard dapat diakses menggunakan Virtual IP yang sudah diset sebelumnya yaitu : http://10.65.65.254



Password dapat dilihat menggunakan command berikut ini:

```
# grep keystone_admin_password
/etc/kolla/passwords.yml
```

Troubleshoot

Ketika host mengalami restart:

```
export EXT_NET_CIDR='10.25.25.0/24'
export EXT_NET_RANGE='start=10.25.25.3,end=10.25.25.253'
export EXT_NET_GATEWAY='10.25.25.1'
sudo ifconfig br-ex $EXT_NET_GATEWAY netmask 255.255.255.0
up
sudo iptables -t nat -A POSTROUTING -s $EXT_NET_CIDR -o
ens224 -j MASQUERADE
echo 1 > /proc/sys/net/ipv4/ip_forward
sudo iptables -A FORWARD -o ens224 -i br-ex -j ACCEPT
sudo iptables -A FORWARD -i ens224 -o br-ex -j ACCEPT
```

Membuat Instance Melalui Horizon

- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

- ✓ **Openstack: Upload Image atau ISO**
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

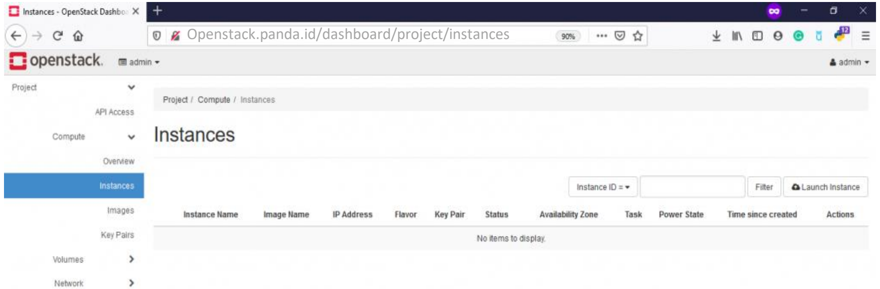
✓ Upload Image atau ISO

Create Instance

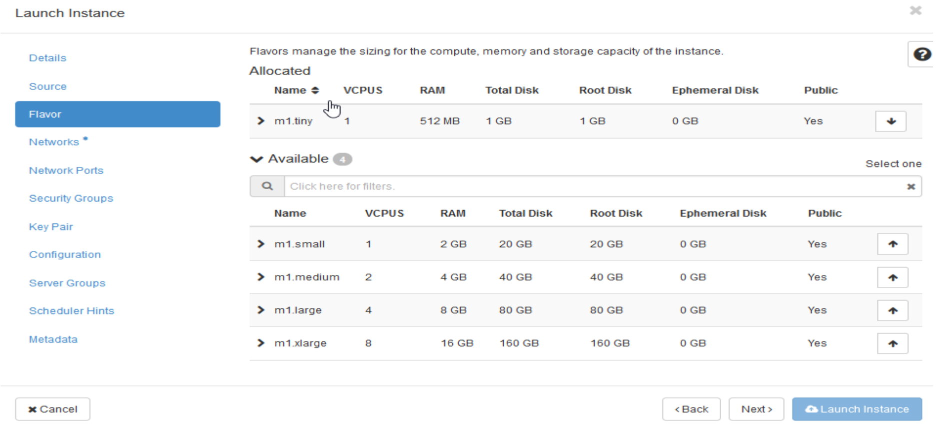
Upload Image atau ISO

Create Instance

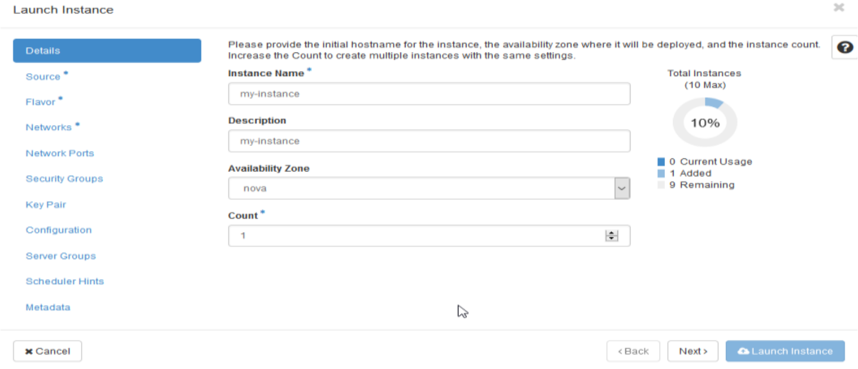
Upload image atau iso menggunakan cirros, silakan Login Horizon >> Project >> Compute >> Instances >> Launch Instance



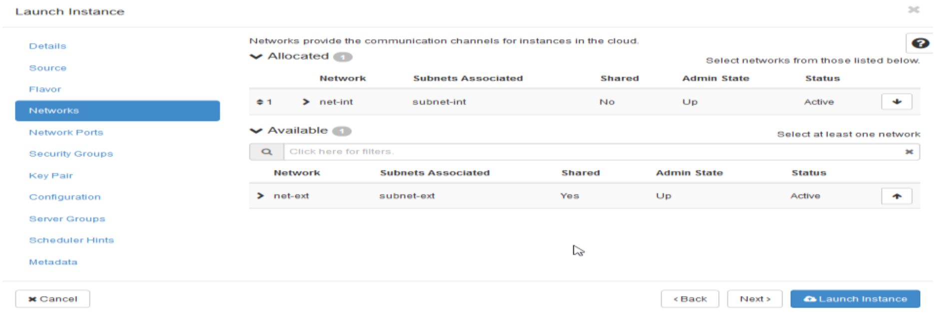
Menu **Flavor** silakan pilih spesifikasi instance



Pada menu Source silakan pilih OS atau image yang ingin Anda gunakan, disini kami contohkan OS yang kami gunakan yaitu Cirros



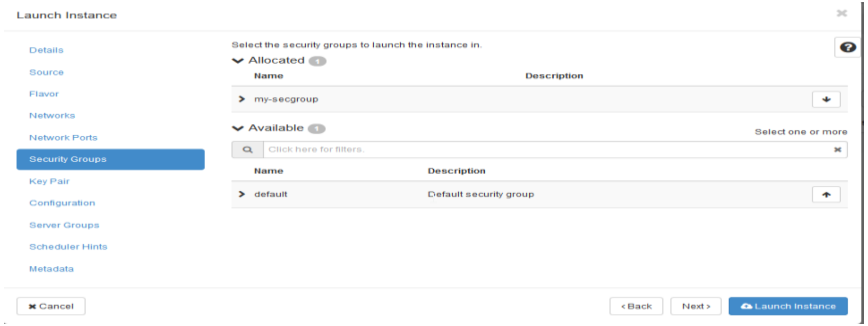
Menu **Networks** silakan pilih network internal yang telah dibuat sebelumnya.



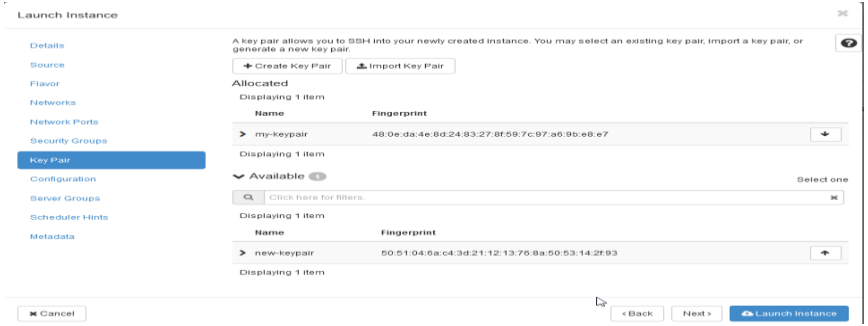
Upload Image atau ISO

Create Instance

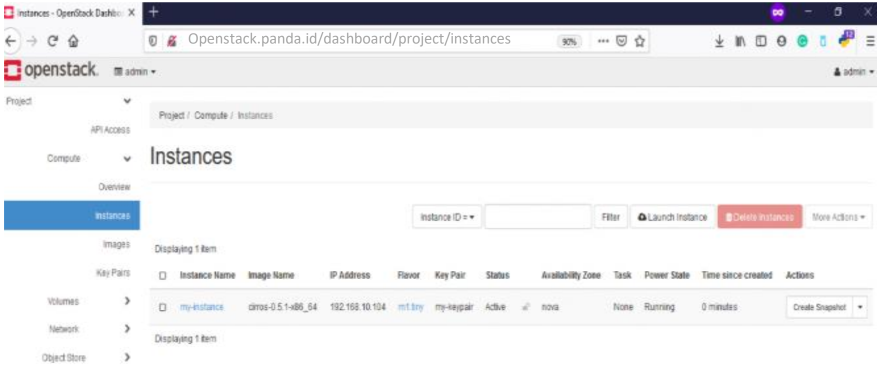
Menu Security Groups , silakan pilih Security Groups yang telah dibuat sebelumnya.



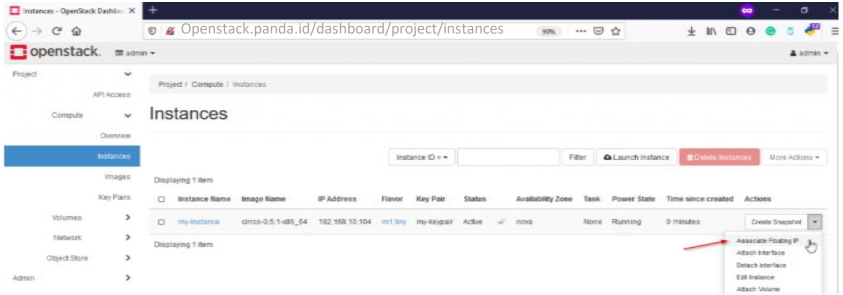
Menu **Key Pair** silakan pilih key pair yang telah di buat juga sebelumnya



Jika sudah silakan, create **Launch Instance** silakan tunggu proses pembuatan Instance selesai



Saat ini instance sudah mendapatkan IP internal secara otomatis, supaya dapat diakses dari public silakan membuat floating IP pada instance silakan klik menu Associate Floating IP

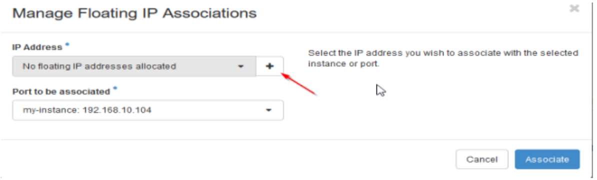


Klik tanda +

Lab: Kolla Ansible All in One Openstack (Cons't)

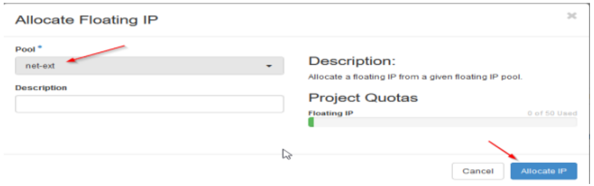
Upload Image atau ISO
Create Instance

Manage Floating IP Associations

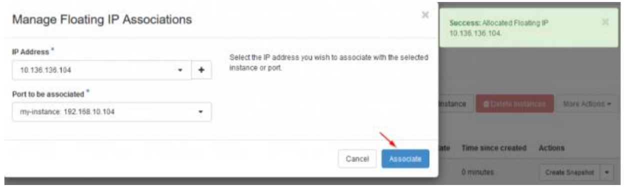


Pilih network external >> Allocate IP

Pilih network external >> Allocate IP



Saat ini kita sudah mendapatkan IP nya, silakan klik **Associate** untuk dapat di assign ke instance Anda

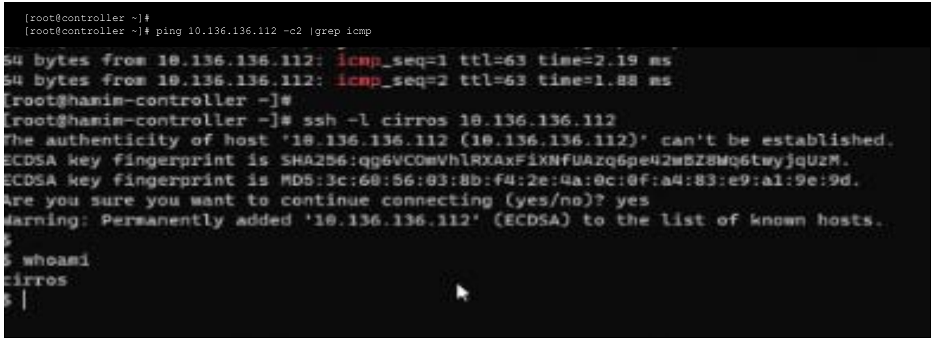


Saat ini instance sudah mendapatkan dan menggunakan IP dari network internal dan external dan silakan test ping ke IP floating

```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c4
```

Test login ke sisi instance

```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c2 |grep icmp
```



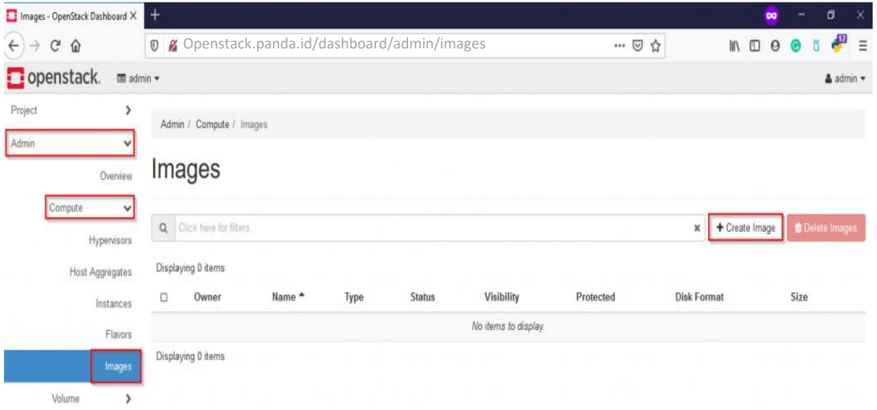
Upload Image atau ISO

Create Image

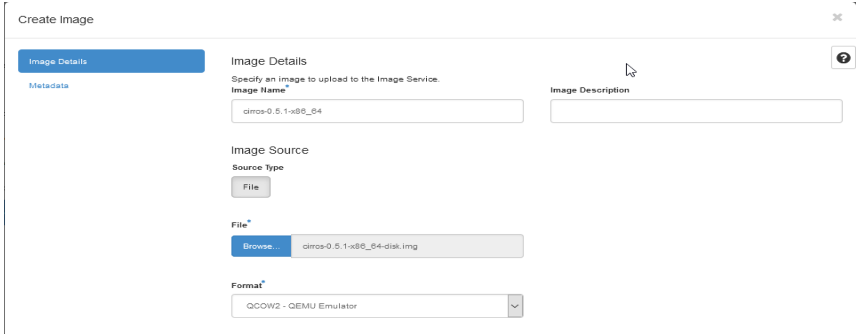
upload image atau iso di openstack terdapat 2 cara yang dapat dilakukan yaitu melalui horizon (melalui web) atau CLI.
Upload image atau iso pastikan Anda sudah mengunguh image nya terlebih dahulu berikut link jika menggunakan Ubuntu, CentOS dan Cirros

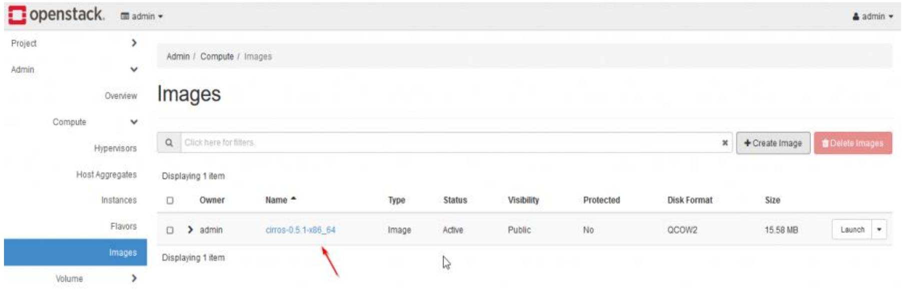
- Ubuntu Cloud Images
- CentOS Cloud Images
- Cirros

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



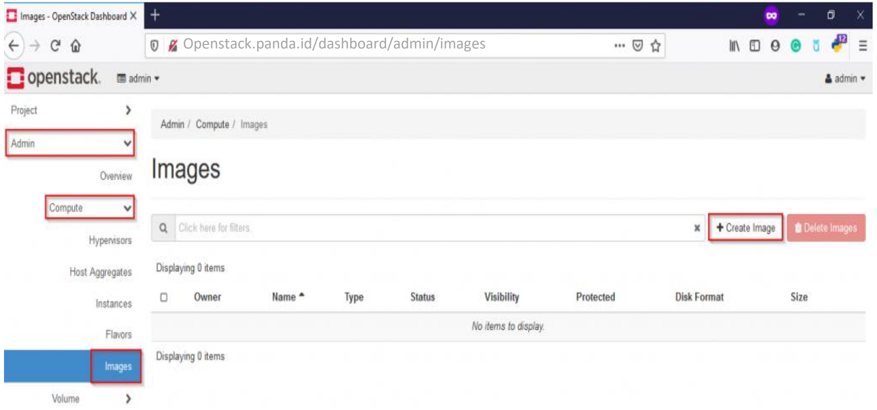
Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*



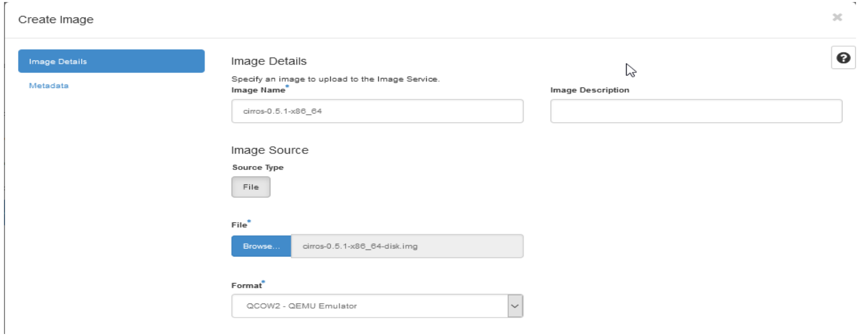


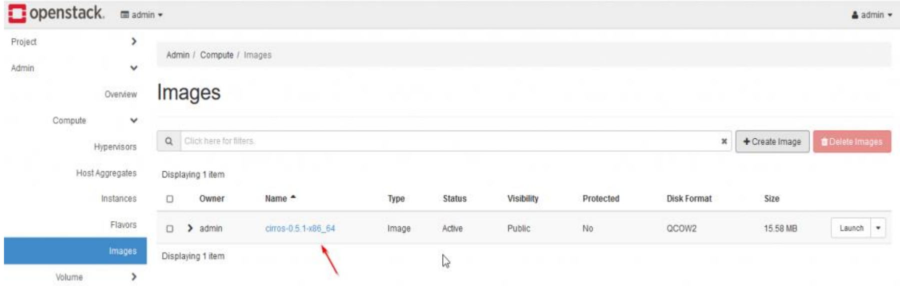
Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



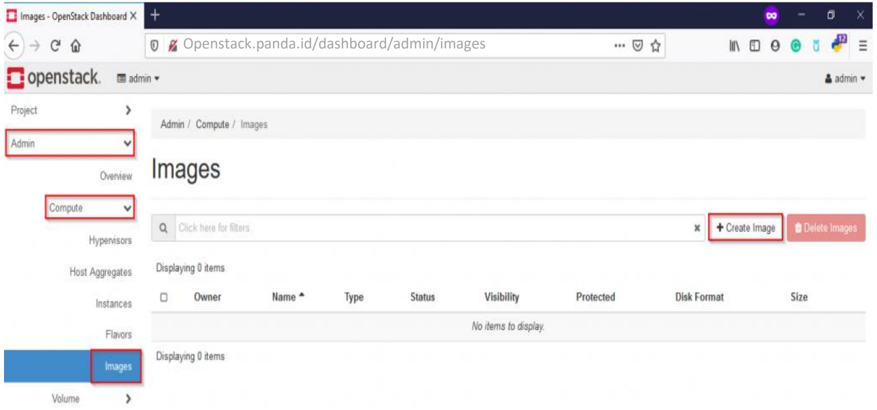
Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*



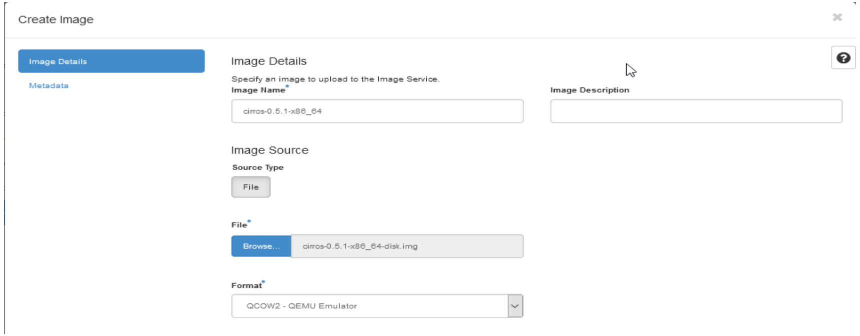


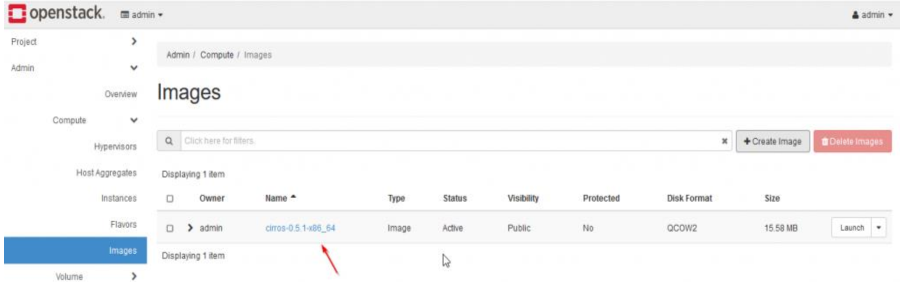
Create Instance Melalui Horizon

Upload image atau iso menggunakan cirros, silakan login Horizon >> Admin >> Compute >> Images >> Create Images



Selanjutnya, isi *Image Name* dengan nama image yang Anda inginkan dan klik *Browser >> Upload Image >> Format (QCOW2 – QEMU Emulator) >> Klik Create Image*





- ✓ **Openstack: Upload Image atau ISO**
- ✓ **Openstack: Membuat Network**
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

- ✓ Membuat Network

Create Network

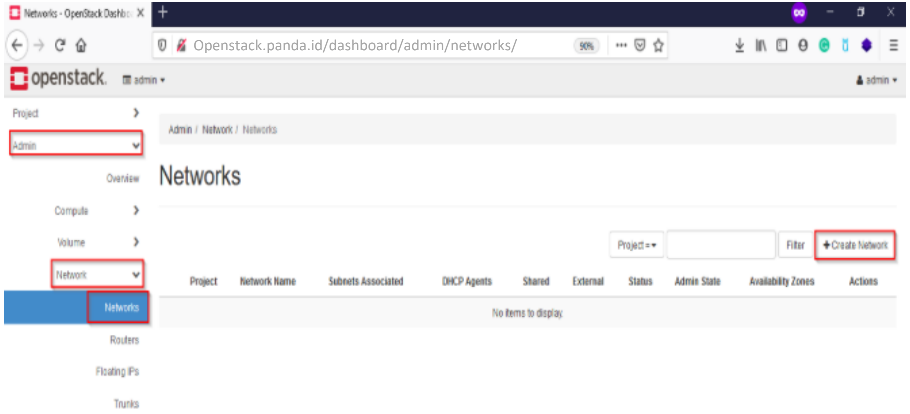
Create Network

Network yaitu network external dan network internal

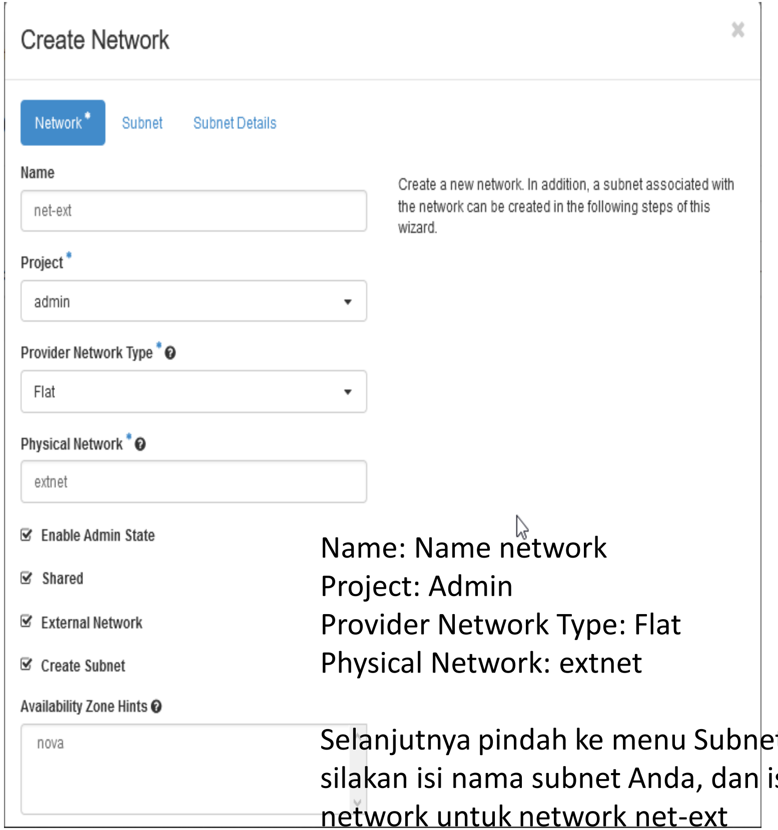
- Network External: Network yang tersambung keluar (node controller dan compute).
- Network internal: Network yang akan terhubung dan digunakan oleh instance/vm.

Login Horizon >> Admin >> Network >> Networks >> Create Network

Openstack.panda.id/dashboard/admin/networks/



Menu ***Network***

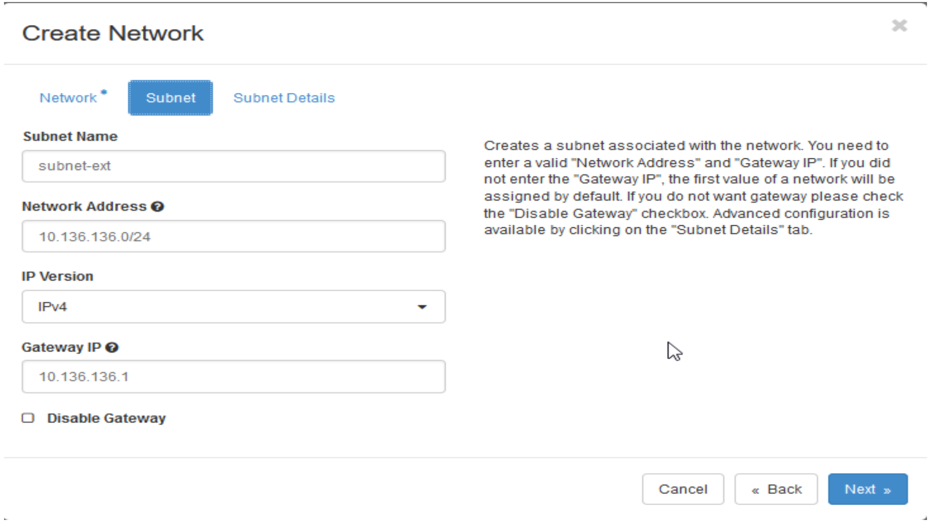


Name: Name network
Project: Admin
Provider Network Type: Flat
Physical Network: extnet

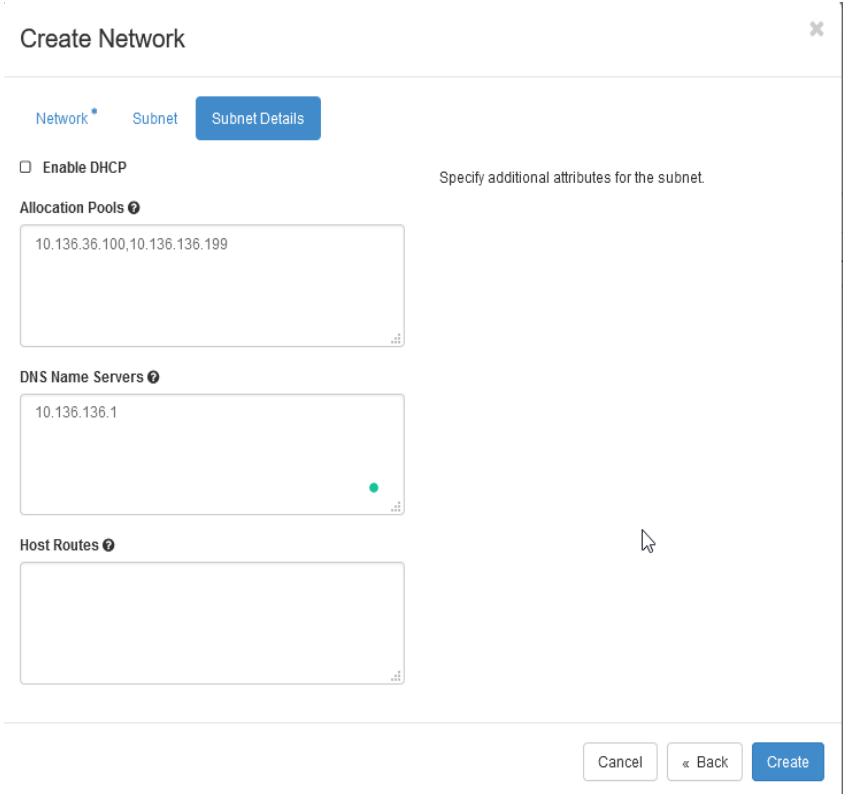
Selanjutnya pindah ke menu Subnet , silakan isi nama subnet Anda, dan isi juga network untuk network net-ext

Create Network

Menu Subnet , silakan isi nama subnet Anda, dan isi juga network untuk network net-ext



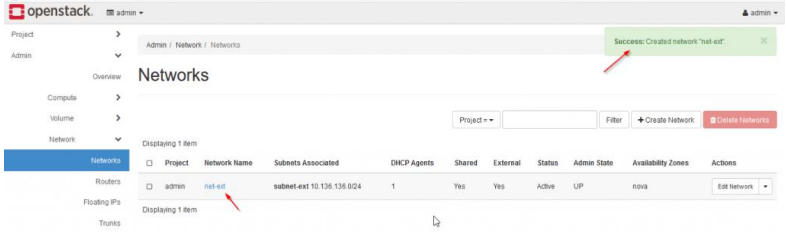
Menu _ Subnet Details _ silakan isi ip pool yang ingin Anda gunakan dari network address diatas, jika sudah silakan klik Create



Lab: Kolla Ansible All in One Openstack (Cons't)

Create Network

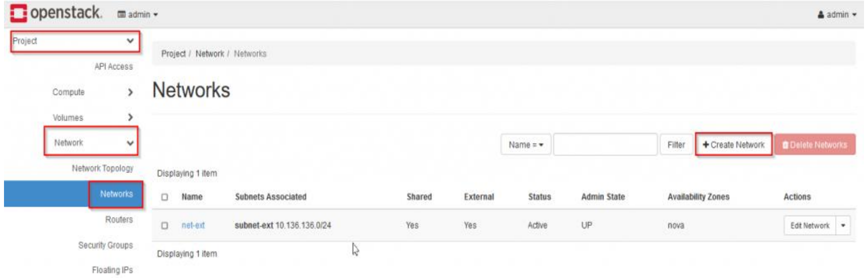
Pembuatan network selesai



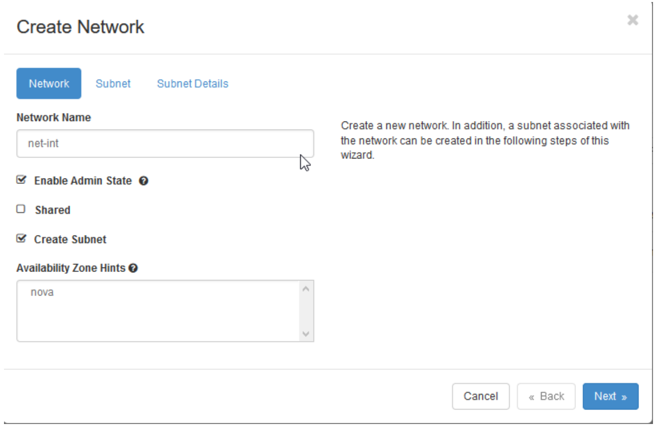
Saat ini external network sudah berhasil dibuat, selanjutnya create internal network seperti berikut:

Login Horizon >> Project >> Network >> Networks >> Create Network

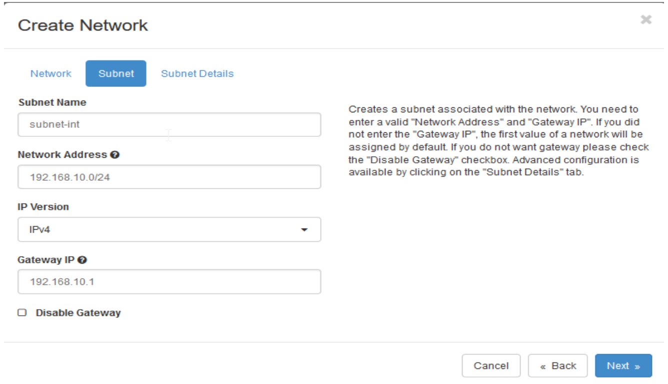
Openstack.panda.id/dashboard/admin/networks/Create Network



Menu Network isikan nama network

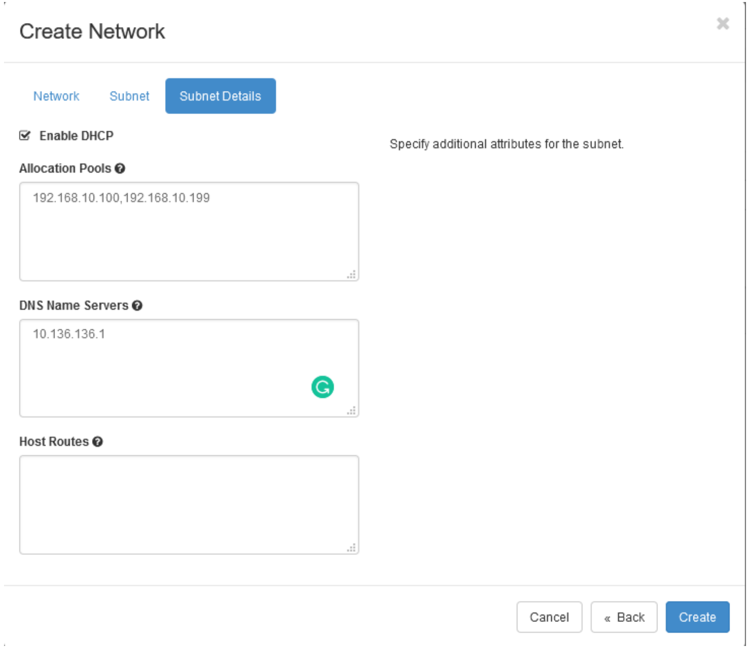


Pindah ke menu subnet untuk menentukan ip address dan subnet nya

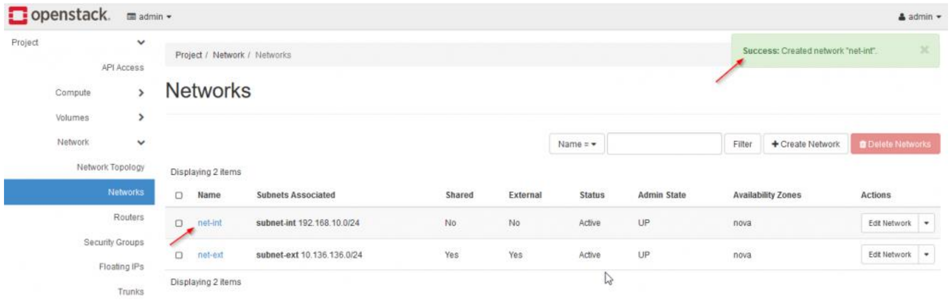


Create Network

Pada menu Subnet Details, silakan isi pool network yang ingin digunakan dari range ip address diatas lalu klik _ Create _ untuk membuat network internal



Proses pembuatan network internal



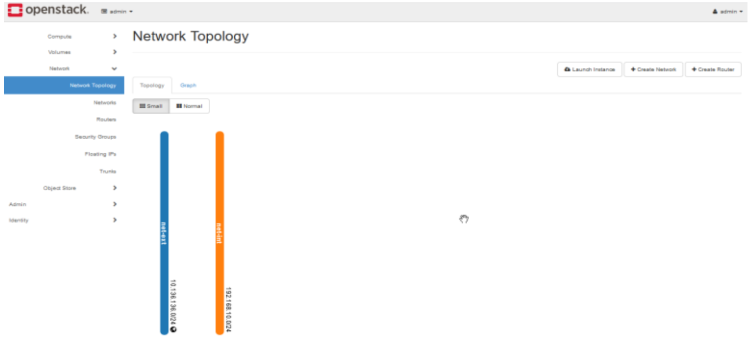
- ✓ **Openstack: Upload Image atau ISO**
- ✓ **Openstack: Membuat Network**
- ✓ **Openstack: Membuat Router**
- ✓ Openstack: Menambahkan SSH Key
- ✓ Openstack: Menambah dan Menggunakan Security Group

✓ Membuat Router

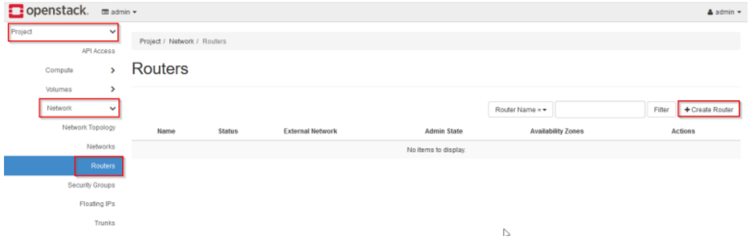
Create Router

Create Router

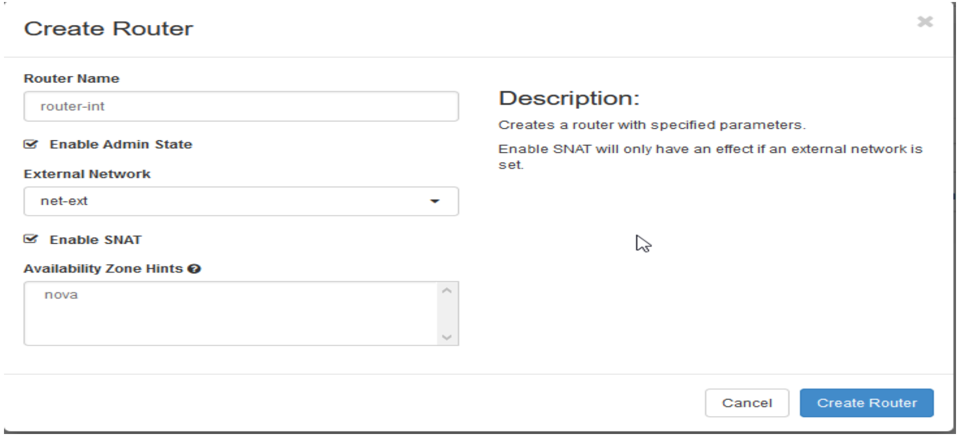
Membuat router melalui horizon, fungsi router sendiri untuk menghubungkan network yang telah kita buat sebelumnya, jika di lihat melalui *Network topologi* saat ini sudah terdapat 2 network internal dan external

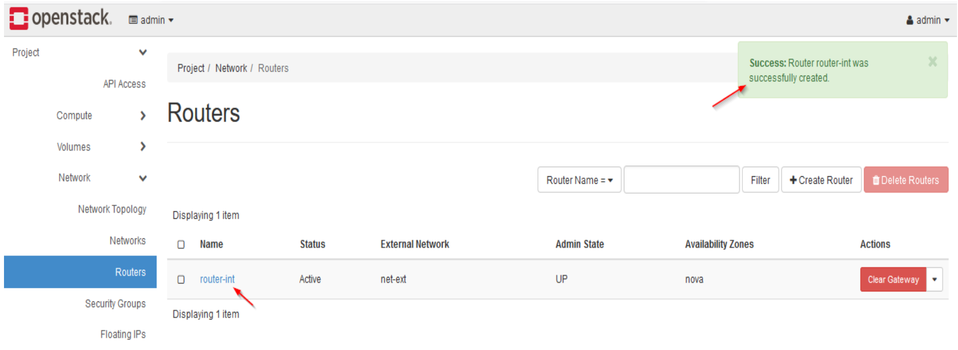


Membuat router Login ke Horizon >> Network >> Router >> Create Router



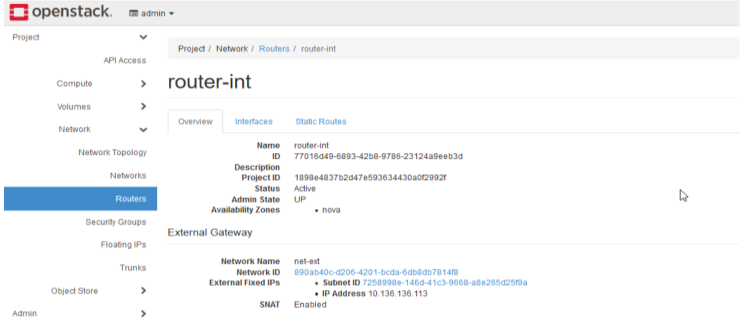
Isi nama Router yang Anda inginkan contoh seperti gambar dibawah ini, klik **Create Router** untuk membuat router



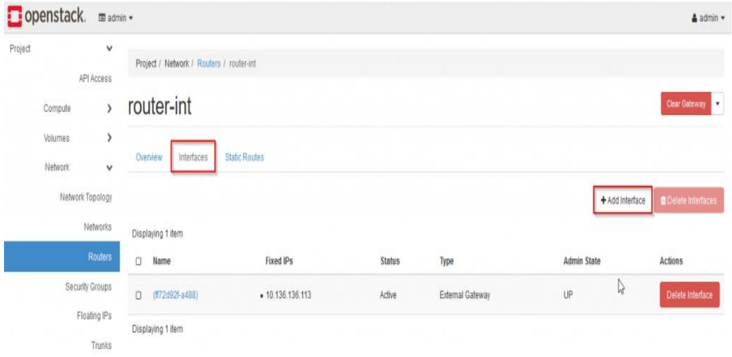


Create Router

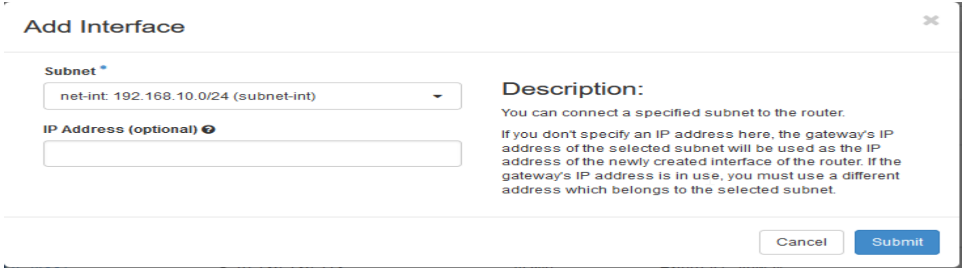
Detail informasi dari router



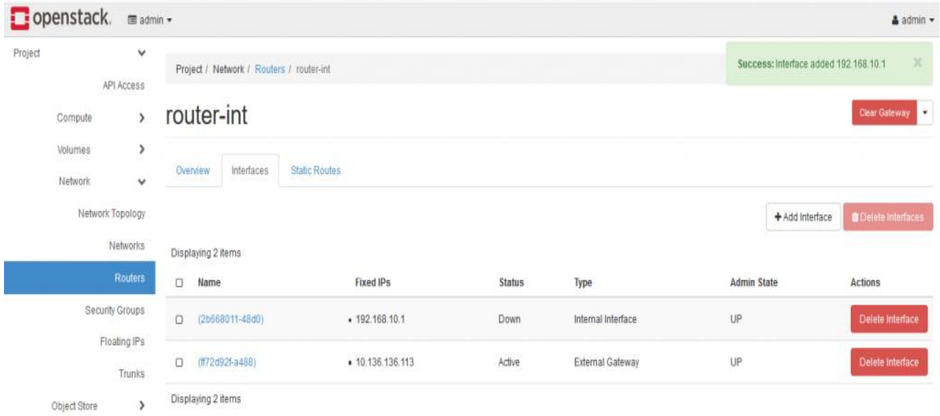
Selanjutnya pindah ke menu **Interface >> Add Interface** untuk menambahkan interface network yang sudah kita buat sebelumnya



Pilih network internal yang sudah dibuat sebelumnya lalu **Submit**



Saat ini interface sudah berhasil di tambahkan

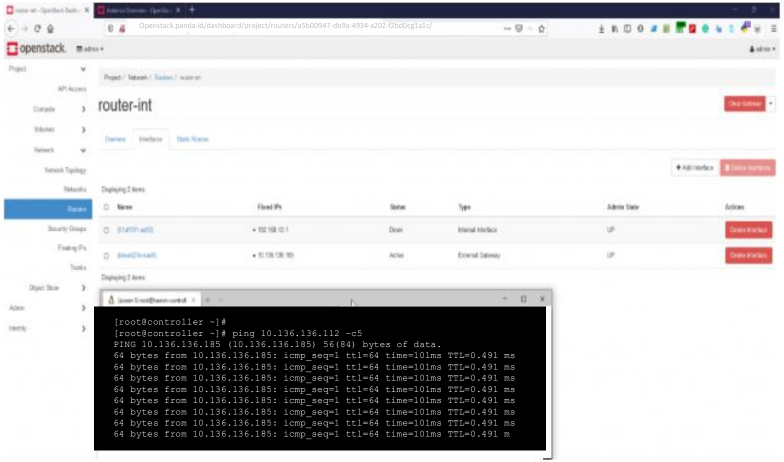


Create Router

Proses pembuatan network internal

Selajutnya test ping dari controller ke IP Gateway atau IP Router yang sudah kita buat sebelumnya.

Openstack.panda.id/dashboard/project/routers/a5b00947-db9a-4934-a202-f2bd0cg1a1c/




```
[root@controller ~]#
[root@controller ~]# ping 10.136.136.112 -c5
PING 10.136.136.185 (10.136.136.185) 56(84) bytes of data.
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
64 bytes from 10.136.136.185: icmp_seq=1 ttl=64 time=101ms TTL=0.491 ms
```

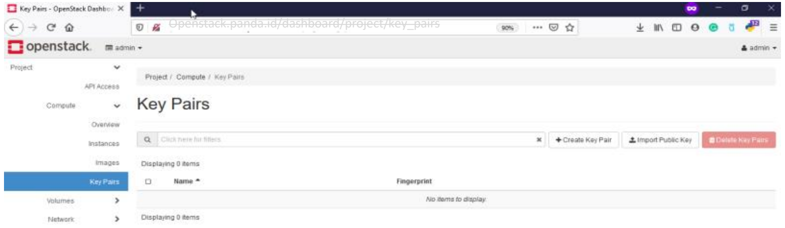
✓ Menambahkan SSH Key

Create SSH keys

Create SSH Keys

Menambahkan SSH Key di openstack. Di openstack untuk menambahkan SSH Key dapat melalui CLI ataupun Horizon, disini kami contohkan menggunakan Horizon.

Login ke *Horizon >> Compute >> Key Pairs*
Openstack.panda.id/dashboard/project/key_pairs



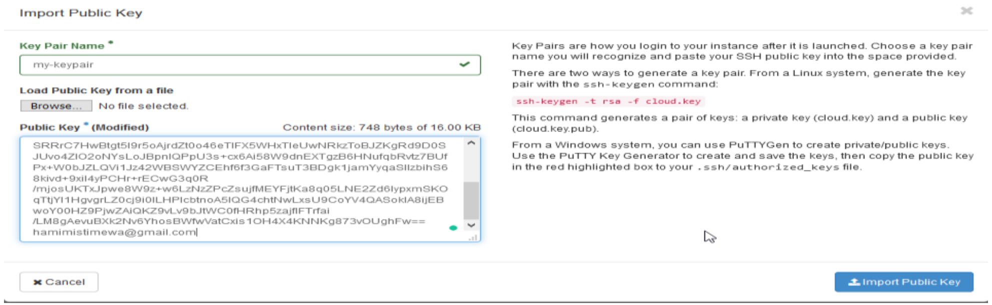
Ada 2 pemilihan terkait SSH Key diantaranya:

a) Import Public Key: Jika Anda sudah mempunyai public key pribadi Anda dapat upload public key nya tanpa create key baru.
b) Create Key Pair: Anda dapat membuat keypair baru dan keypair ini akan Anda gunakan untuk login ke instance

Berikut kami berikan contoh masing – masing pilihan diatas: Import Public Key.

✓ ***Import Public Key***

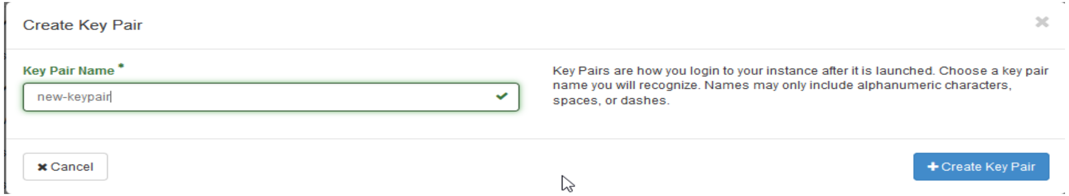
Silakan klik menu **Import Public Key** lalu berikan nama key Anda dan isi public key Anda contoh nya seperti gambar dibawah lalu klik Import Public Key



Create SSH Keys

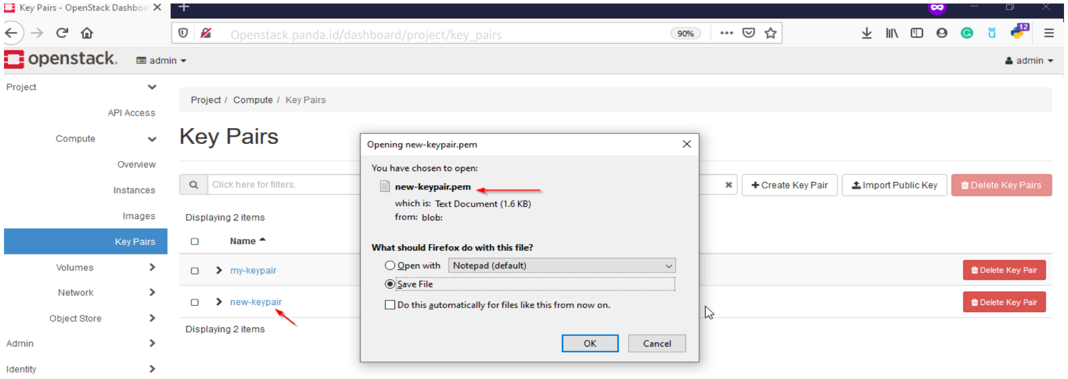
Create Key Pair

Silakan Klik pada **Create Key Pair** lalu isi nama keypair



silakan klik Create Key Pair, setelah nya akan muncul pop seperti Anda, silakan download key pair nya.

Openstack.panda.id/dashboard/project/key_pairs



- ✓ Openstack: Upload Image atau ISO
- ✓ Openstack: Membuat Network
- ✓ Openstack: Membuat Router
- ✓ Openstack: Menambahkan SSH Key
- ✓ **Openstack: Menambah dan Menggunakan Security Group**

✓ Menambah Security

Create Security Group

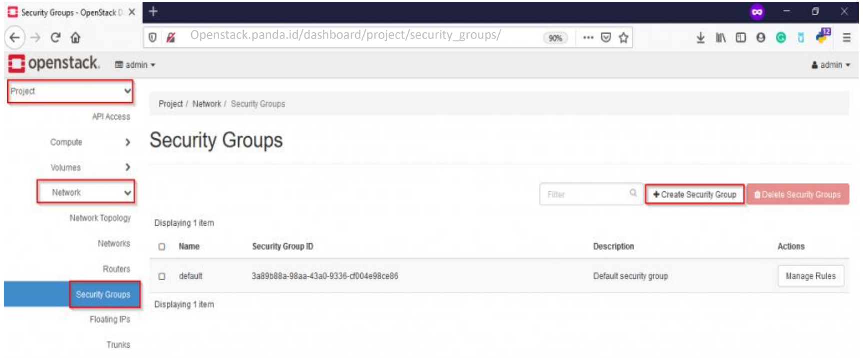
Lab: Kolla Ansible All in One Openstack (Cons't)

Create Security Group

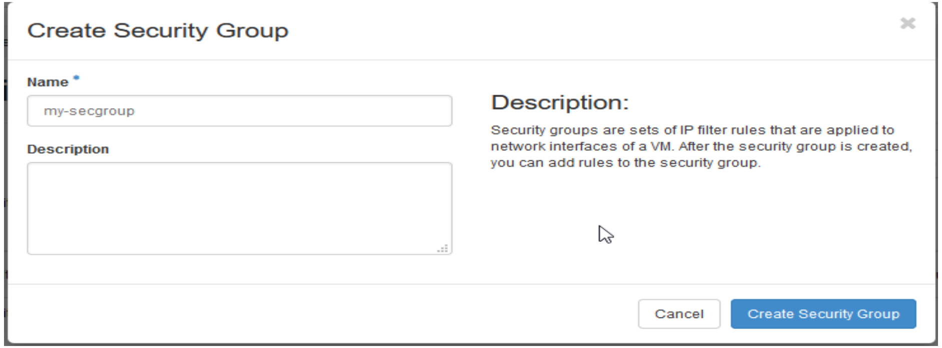
Membuat dan menggunakan security group dapat melalui horizon atau via CLI. Disini kami akan menggunakan horizon untuk menambahkan security group, berikut tahapannya:

Login Horizon >> Project >> Network >> Security Group >> +Create Security Group

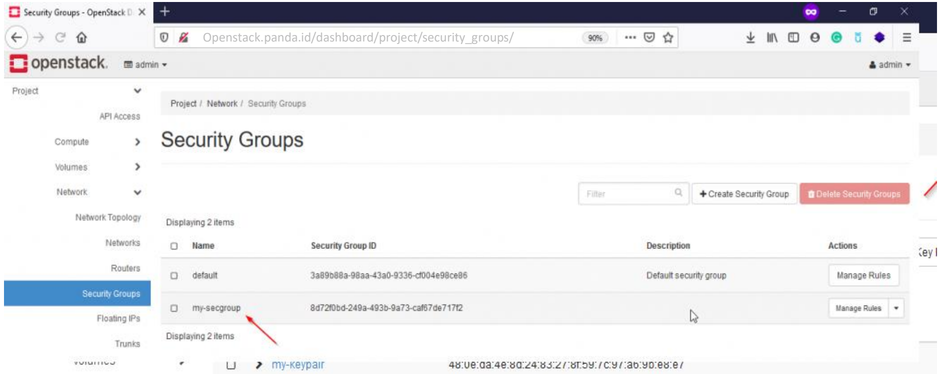
Openstack.panda.id/dashboard/project/security_groups/



Isi nama security group Anda lalu klik Create Security Group



Saat ini security grouup sudah berhasil

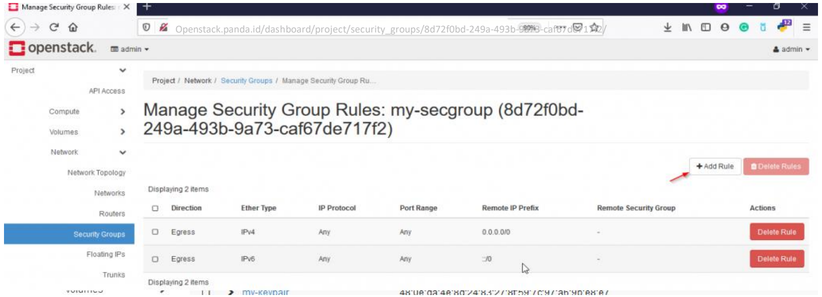


Lab: Kolla Ansible All in One Openstack (Cons’t)

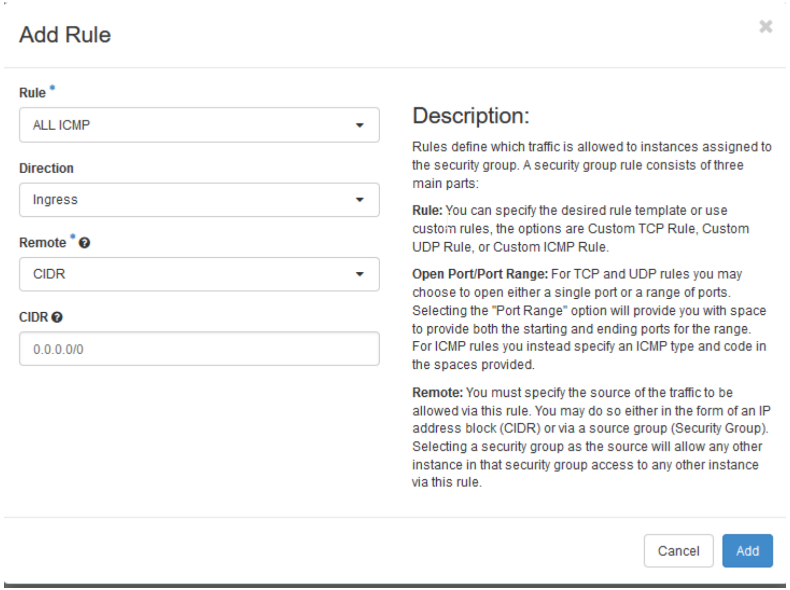
Create Security Group

Terdapat 2 security group (default) dan security group yang sudah dibuat sebelumnya.
Untuk yang default sebenarnya dapat Anda gunakan juga. Untuk menambahkan port atau menggunakan security group silakan klik menu **Manage Rules** pada security group Anda

Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/

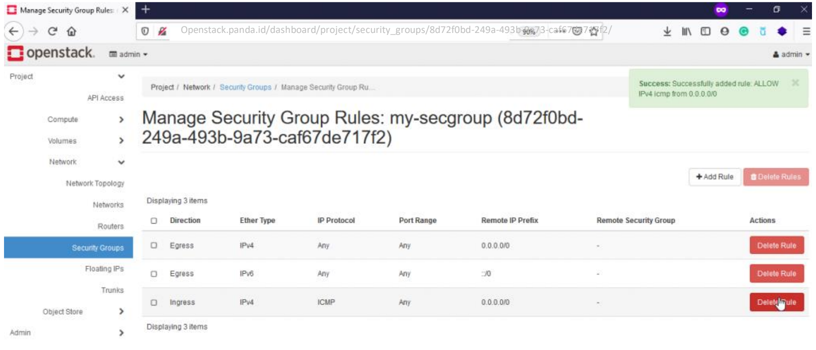


Klik + Add Rule untuk menambahkan rule atau port yang ingin Anda gunakan, misalnya disini kami ingin allow port atau protokol IMCP, lalu klik Add



Create SSH Keys

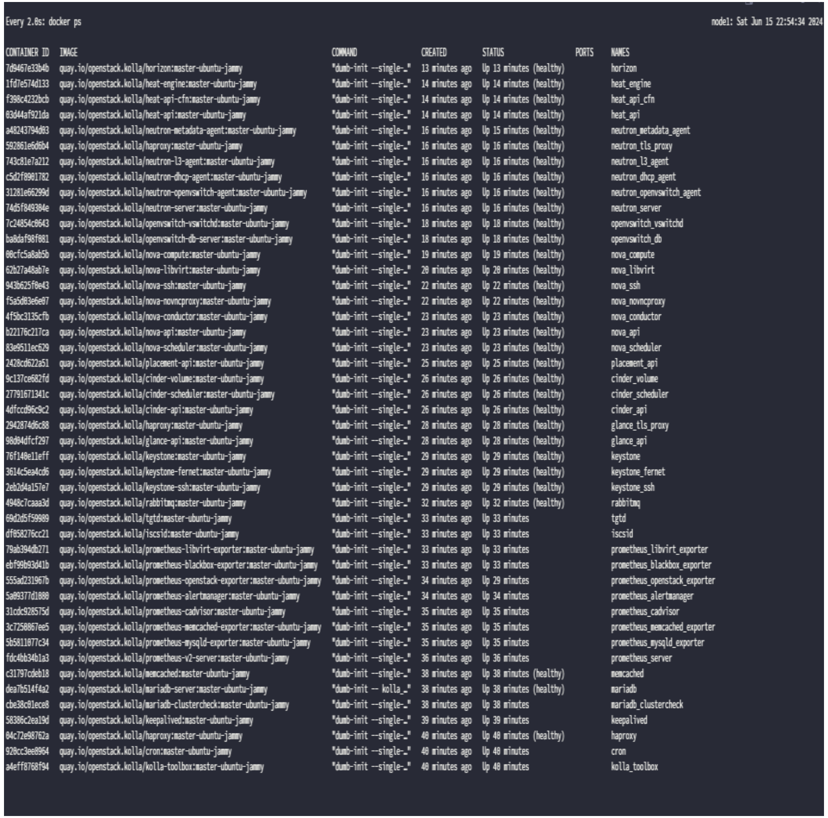
Saat ini rule ICMP sudah dibuat
Openstack.panda.id/dashboard/project/security_groups/8d72f0bd-249a-493b-9a73-caf67de717f2/



Dapat menambahkan rule atau port – port yang Anda inginkan misalnya port ssh, imap, http dan yang lainnya melalui security group

Lab: Kolla Ansible All in One Openstack (Cons’t)

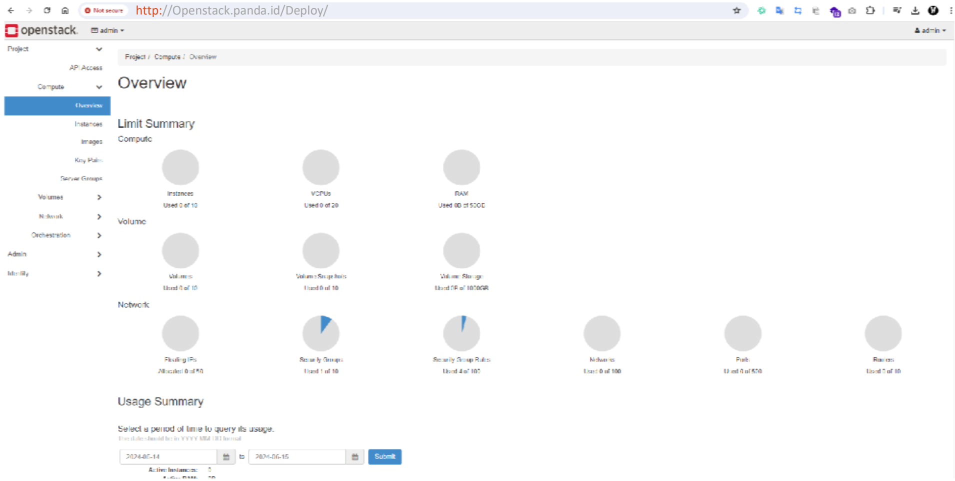
All Service running on Docker



```
Every 2.0s: docker ps                                                                                      node1: Sat Jun 15 22:54:34 2024

CONTAINER ID   IMAGE                                                             COMMAND                  CREATED          STATUS                    PORTS     NAMES
7d9467e33b4b   quay.io/openstack.kolla/horizon:master-ubuntu-jammy               "dumb-init --single-…"   13 minutes ago   Up 13 minutes (healthy)             horizon
1fd7e574d133   quay.io/openstack.kolla/heat-engine:master-ubuntu-jammy           "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_engine
f398c4232bcb   quay.io/openstack.kolla/heat-api-cfn:master-ubuntu-jammy          "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_api_cfn
03d44af921da   quay.io/openstack.kolla/heat-api:master-ubuntu-jammy              "dumb-init --single-…"   14 minutes ago   Up 14 minutes (healthy)             heat_api
a48243794d03   quay.io/openstack.kolla/neutron-metadata-agent:master-ubuntu-jammy "dumb-init --single-…"  16 minutes ago   Up 15 minutes (healthy)             neutron_metadata_agent
592861e6d6b4   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy               "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_tls_proxy
743c81e7a212   quay.io/openstack.kolla/neutron-l3-agent:master-ubuntu-jammy      "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_l3_agent
c5d2f8901782   quay.io/openstack.kolla/neutron-dhcp-agent:master-ubuntu-jammy    "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_dhcp_agent
31281e66299d   quay.io/openstack.kolla/neutron-openvswitch-agent:master-ubuntu-jammy "dumb-init --single-…" 16 minutes ago Up 16 minutes (healthy)           neutron_openvswitch_agent
74d5f849384e   quay.io/openstack.kolla/neutron-server:master-ubuntu-jammy        "dumb-init --single-…"   16 minutes ago   Up 16 minutes (healthy)             neutron_server
7c24854c0643   quay.io/openstack.kolla/openvswitch-vswitchd:master-ubuntu-jammy  "dumb-init --single-…"   18 minutes ago   Up 18 minutes (healthy)             openvswitch_vswitchd
ba8daf98f081   quay.io/openstack.kolla/openvswitch-db-server:master-ubuntu-jammy "dumb-init --single-…"   18 minutes ago   Up 18 minutes (healthy)             openvswitch_db
00cfc5a8ab5b   quay.io/openstack.kolla/nova-compute:master-ubuntu-jammy          "dumb-init --single-…"   19 minutes ago   Up 19 minutes (healthy)             nova_compute
62b27a48ab7e   quay.io/openstack.kolla/nova-libvirt:master-ubuntu-jammy          "dumb-init --single-…"   20 minutes ago   Up 20 minutes (healthy)             nova_libvirt
943b625f0e43   quay.io/openstack.kolla/nova-ssh:master-ubuntu-jammy              "dumb-init --single-…"   22 minutes ago   Up 22 minutes (healthy)             nova_ssh
f5a5d03e6e07   quay.io/openstack.kolla/nova-novncproxy:master-ubuntu-jammy       "dumb-init --single-…"   22 minutes ago   Up 22 minutes (healthy)             nova_novncproxy
4f5bc3135cfb   quay.io/openstack.kolla/nova-conductor:master-ubuntu-jammy        "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_conductor
b22176c217ca   quay.io/openstack.kolla/nova-api:master-ubuntu-jammy              "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_api
83e9511ec629   quay.io/openstack.kolla/nova-scheduler:master-ubuntu-jammy        "dumb-init --single-…"   23 minutes ago   Up 23 minutes (healthy)             nova_scheduler
2428cd622a51   quay.io/openstack.kolla/placement-api:master-ubuntu-jammy         "dumb-init --single-…"   25 minutes ago   Up 25 minutes (healthy)             placement_api
9c137ce682fd   quay.io/openstack.kolla/cinder-volume:master-ubuntu-jammy         "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_volume
27791671341c   quay.io/openstack.kolla/cinder-scheduler:master-ubuntu-jammy      "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_scheduler
4dfccd96c9c2   quay.io/openstack.kolla/cinder-api:master-ubuntu-jammy            "dumb-init --single-…"   26 minutes ago   Up 26 minutes (healthy)             cinder_api
2942874d6c88   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy               "dumb-init --single-…"   28 minutes ago   Up 28 minutes (healthy)             glance_tls_proxy
98d04dfcf297   quay.io/openstack.kolla/glance-api:master-ubuntu-jammy            "dumb-init --single-…"   28 minutes ago   Up 28 minutes (healthy)             glance_api
76f140e11eff   quay.io/openstack.kolla/keystone:master-ubuntu-jammy              "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone
3614c5ea4cd6   quay.io/openstack.kolla/keystone-fernet:master-ubuntu-jammy       "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone_fernet
2eb2d4a157e7   quay.io/openstack.kolla/keystone-ssh:master-ubuntu-jammy          "dumb-init --single-…"   29 minutes ago   Up 29 minutes (healthy)             keystone_ssh
4948c7caaa3d   quay.io/openstack.kolla/rabbitmq:master-ubuntu-jammy              "dumb-init --single-…"   32 minutes ago   Up 32 minutes (healthy)             rabbitmq
69d2d5f59989   quay.io/openstack.kolla/tgtd:master-ubuntu-jammy                  "dumb-init --single-…"   33 minutes ago   Up 33 minutes                       tgtd
df058276cc21   quay.io/openstack.kolla/iscsid:master-ubuntu-jammy                "dumb-init --single-…"   33 minutes ago   Up 33 minutes                       iscsid
79ab394db271   quay.io/openstack.kolla/prometheus-libvirt-exporter:master-ubuntu-jammy "dumb-init --single-…" 33 minutes ago Up 33 minutes                     prometheus_libvirt_exporter
ebf99b93d41b   quay.io/openstack.kolla/prometheus-blackbox-exporter:master-ubuntu-jammy "dumb-init --single-…" 33 minutes ago Up 33 minutes                    prometheus_blackbox_exporter
555ad231967b   quay.io/openstack.kolla/prometheus-openstack-exporter:master-ubuntu-jammy "dumb-init --single-…" 34 minutes ago Up 29 minutes                   prometheus_openstack_exporter
5a09377d1880   quay.io/openstack.kolla/prometheus-alertmanager:master-ubuntu-jammy "dumb-init --single-…"  34 minutes ago   Up 34 minutes                       prometheus_alertmanager
31cdc928575d   quay.io/openstack.kolla/prometheus-cadvisor:master-ubuntu-jammy   "dumb-init --single-…"   35 minutes ago   Up 35 minutes                       prometheus_cadvisor
3c7258867ee5   quay.io/openstack.kolla/prometheus-memcached-exporter:master-ubuntu-jammy "dumb-init --single-…" 35 minutes ago Up 35 minutes                   prometheus_memcached_exporter
5b5811077c34   quay.io/openstack.kolla/prometheus-mysqld-exporter:master-ubuntu-jammy "dumb-init --single-…" 35 minutes ago Up 35 minutes                      prometheus_mysqld_exporter
fdc4bb34b1a3   quay.io/openstack.kolla/prometheus-v2-server:master-ubuntu-jammy  "dumb-init --single-…"   36 minutes ago   Up 36 minutes                       prometheus_server
c31797cdeb18   quay.io/openstack.kolla/memcached:master-ubuntu-jammy             "dumb-init --single-…"   38 minutes ago   Up 38 minutes (healthy)             memcached
dea7b514f4a2   quay.io/openstack.kolla/mariadb-server:master-ubuntu-jammy        "dumb-init -- kolla_…"   38 minutes ago   Up 38 minutes (healthy)             mariadb
cbe38c01ece8   quay.io/openstack.kolla/mariadb-clustercheck:master-ubuntu-jammy  "dumb-init --single-…"   38 minutes ago   Up 38 minutes                       mariadb_clustercheck
58386c2ea19d   quay.io/openstack.kolla/keepalived:master-ubuntu-jammy            "dumb-init --single-…"   39 minutes ago   Up 39 minutes                       keepalived
04c72e98762a   quay.io/openstack.kolla/haproxy:master-ubuntu-jammy               "dumb-init --single-…"   40 minutes ago   Up 40 minutes (healthy)             haproxy
920cc3ee8964   quay.io/openstack.kolla/cron:master-ubuntu-jammy                  "dumb-init --single-…"   40 minutes ago   Up 40 minutes                       cron
a4eff8768f94   quay.io/openstack.kolla/kolla-toolbox:master-ubuntu-jammy         "dumb-init --single-…"   40 minutes ago   Up 40 minutes                       kolla_toolbox
```

Horizon



Lab: Kolla Ansible All in One Openstack (Cons't)

Network Agent

```
root@node1:~# openstack network agent list
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host  | Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
| 53099b35-21c1-4740-acd6-34781fbbc720 | Open vSwitch agent | node1 | None              | :-)   | UP    | neutron-openvswitch-agent |
| 6605786d-3110-43ee-b526-741b2b43abb9 | L3 agent           | node1 | nova              | :-)   | UP    | neutron-l3-agent          |
| 8ef397db-e6c9-418a-a532-dc6ccc330d97 | DHCP agent         | node1 | nova              | :-)   | UP    | neutron-dhcp-agent        |
| b3c9da4e-d28e-4c51-b767-c4c8e4d208da | Metadata agent     | node1 | None              | :-)   | UP    | neutron-metadata-agent    |
+--------------------------------------+--------------------+-------+-------------------+-------+-------+---------------------------+
```

Compute Service

```
root@node1:~# openstack compute service list
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
| ID                                   | Binary         | Host  | Zone     | Status  | State | Updated At                 |
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
| 058a6902-5f6c-4907-ad8e-84ec76c75f69 | nova-scheduler | node1 | internal | enabled | up    | 2024-06-15T15:57:17.000000 |
| 7c2550c7-799b-495e-ac1a-71bcab069b41 | nova-conductor | node1 | internal | enabled | up    | 2024-06-15T15:57:13.000000 |
| b4998b8f-c07a-4b0e-9330-df2b1a9b0206 | nova-compute   | node1 | nova     | enabled | up    | 2024-06-15T15:57:11.000000 |
+--------------------------------------+----------------+-------+----------+---------+-------+----------------------------+
root@node1:~#
```

Volume Service

```
root@node1:~# openstack volume service list
+------------------+-------------+------+---------+-------+----------------------------+
| Binary           | Host        | Zone | Status  | State | Updated At                 |
+------------------+-------------+------+---------+-------+----------------------------+
| cinder-scheduler | node1       | nova | enabled | up    | 2024-06-15T15:59:53.000000 |
| cinder-volume    | node1@lvm-1 | nova | enabled | up    | 2024-06-15T15:59:44.000000 |
+------------------+-------------+------+---------+-------+----------------------------+
root@node1:~#
```

Compute Service

Openstack.panda.id/dashboard/admin/images

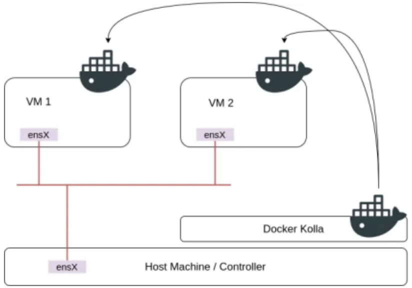
Openstack.panda.id/Deploy/

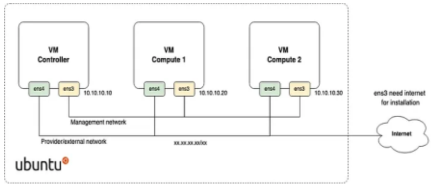
Deploy All-in-One OpenStack with Kolla-Ansible on Ubuntu 22.04

OpenStack Series #2 — Install openstack menggunakan Kolla Ansible (Multi-nodes)

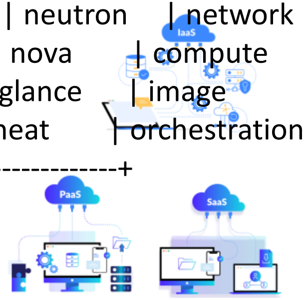
Prasyarat

1. Bare-metal / Virtual Machine (Minimal 1 node untuk metode All In One dan minimal 2 node untuk metode Multinode)
2. Minimal 2 NIC per VM dengan 1 interface yang tidak diberikan ip
3. Disk tambahan untuk backend storage (At Least 50gb)
4. Ram minimal 8gb
5. Koneksi ke Internet
6. Mempunyai akses root maupun user sudoer






```
$ openstack service list
+----------------------------------+-------------+----------------+
| ID                               | Name        | Type           |
+----------------------------------+-------------+----------------+
| 10cb2de52c404662aa6ca3ccdb8f2cf7 | cinderv2    | volumev2       |
| 250a6a7cf4d44f0c9cdd2a39f8203ccc | cinderv3    | volumev3       |
| 2d99d37fc498465591faa9f44580d991 | placement   | placement      |
| 4001de18bf5c45f5a1467a31ea5a2afd | heat-cfn    | cloudformation |
| 505333b0b90b4962a37d55c44bc940f9 | keystone    | identity       |
| 612c645a37e848d49af9f6f53dc6e8ac | nova_legacy | compute_legacy |
| 87c2edd540c84c8296251ea284a14398 | neutron     | network        |
| 94e54356818d43af8ed7f4e1a6da28f1 | nova        | compute        |
| a9f6eb791ec740c183ecff67dc7c7b08 | glance      | image          |
| eaf1f96955244acfbf5124e42c32f4e6 | heat        | orchestration  |
+----------------------------------+-------------+----------------+
```



Openstack.panda.id/dashboard/admin/images

Openstack.panda.id/Deploy/

Openstack 2024.2 with Kolla Ansible

Openstack 2024.2 with Kolla Ansible (Cons’t)

Node Hostname	*Node Role*	*vCPU*	*Memory*	*RootDisk*	*ManagementNet*	*StorageNet*
GPU01adm01	Ansible	2 Core	2GB	20GB	10.78.78.199	-
GPU01con01	Controller	16 Core	16GB	40GB	10.78.78.201	10.79.79.201
GPU01con02	Controller	16 Core	16GB	40GB	10.78.78.202	10.79.79.202
GPU01con03	Controller	16 Core	16GB	40GB	10.78.78.203	10.79.79.203
GPU01hpv01	Hypervisor	32 Core	32GB	40GB	10.78.78.204	10.79.79.204
GPU01hpv02	Hypervisor	32 Core	32GB	40GB	10.78.78.205	10.79.79.205

Openstack 2024.2 with Kolla Ansible (Cons’t)

Interface mapping:

- ✓ eth0 for Public Network
- ✓ eth1 for Management Network
- ✓ eth2 for Storage Network with Jumbo frames
- ✓ eth4 for Tunnel/Self Service/Guest Network with Jumbo frames

We can review network configuration in **sample-netplan.yaml**

```yaml
network:
 version: 2
 ethernets:
  eth0:
   match:
    macaddress: "BC:24:11:D3:02:2F"
   set-name: "eth0"
   critical: true
  eth1:
   match:
    macaddress: "BC:24:11:77:6F:A2"
   set-name: "eth1"
   addresses:
   - "10.78.78.201/24"
   nameservers:
    addresses:
    - 10.78.78.103
    - 10.78.78.104
    search:
    - Deploy.is-a.dev
   routes:
   - to: "default"
    via: "10.78.78.254"
   critical: true
  eth2:
   match:
    macaddress: "BC:24:11:D6:06:52"
   set-name: "eth2"
   addresses:
   - "10.79.79.201/24"
   critical: true
   mtu: 9000
  eth3:
   match:
    macaddress: "BC:24:11:EF:59:FD"
   set-name: "eth3"
   addresses:
   - "10.79.80.201/24"
   critical: true
   mtu: 9000
```

Openstack 2024.2 with Kolla Ansible (Cons't)

Installing NFS Server for Glance Service in GPU01adm01 node

```
sudo apt update
sudo apt install nfs-kernel-server
sudo mkdir -p /data
sudo chown nobody:nogroup /data
echo "/data *(rw,sync,no_subtree_check)" >> /etc/exports
sudo systemctl restart nfs-kernel-server
sudo exportfs -v
```

Execute in all nodes

Mapping static hostname in /etc/hosts

```
cat <<EOF | tee -a /etc/hosts
10.78.78.199 GPU01adm01 GPU01adm01.Deploy.is-a.dev
10.78.78.201 GPU01con01 GPU01con01.Deploy.is-a.dev
10.78.78.202 GPU01con02 GPU01con02.Deploy.is-a.dev
10.78.78.203 GPU01con02 GPU01con03.Deploy.is-a.dev
10.78.78.204 GPU01hpv01 GPU01hpv01.Deploy.is-a.dev
10.78.78.205 GPU01hpv02 GPU01hpv02.Deploy.is-a.dev

10.78.78.200 os-int
10.78.78.100 os-ext
EOF
```

Install packages and dependencies

```
sudo apt install -y git gcc libssl-dev libffi-dev \
 python3-venv python3-dev python3-selinux python3-setuptools \
 python3-pip python3-docker nfs-common
```

Mount nfs volume for all node

```
sudo mkdir /data cat <<EOF | sudo tee -a /etc/fstab 10.78.78.199:/data /data nfs defaults,_netdev 0 0
EOF
 sudo mount /data
```

Create virtual environment for kolla-ansible

```
python3 -m venv kolla-venv
cd kolla-venv
source bin/activate
 pwd
```

```
panda@GPU01adm01: ~ cd kolla-venv
panda@GPU01adm01: ~ /kolla-venv$ source bin/activate
(kola-venv) panda@GPU01adm01: ~ /kolla-venv $ pwd
/home/panda/ kolla-venv
(kola-venv) panda@GPU01adm01: ~ /kolla-venv $
```

Update pip and install ansible

```
pip install -U pip pip install 'ansible>=8,<9'
```

Build Multinode Kolla-Ansible Openstack LAB using LXD

Mauliate Godang